

Code ▼

U4370 HW2

0. Read Packages

Hide

```
library(AER)
library(wooldridge)
library(car)
```

1. Data Preperation

a.

Hide

```
data('mroz')
df <- mroz[mroz$inlf == 1,]
remove(mroz)
head(df,2)
```

	inlf <int>	hours <int>	kidslt6 <int>	kidsge6 <int>	age <int>	educ <int>	wage <dbl>	repwage <dbl>	hushrs <int>
1	1	1610	1	0	32	12	3.3540	2.65	2708
2	1	1656	0	2	30	12	1.3889	2.65	2310

2 rows | 1-10 of 22 columns

b.

Hide

```
paste('sample size = ',dim(df)[1])
```

```
[1] "sample size = 428"
```

2.

Hide

```
vars.Endog <- c('educ')
vars.Exog <- c('exper','expersq',
              'kidslt6','kidsge6',
              'husage','huswage','city')
form.ols <- reformulate(c(vars.Endog,vars.Exog),'lwage')
fit.ols <- lm(form.ols,data = df)
beta.ols <- coef(fit.ols)[vars.Endog]
beta.ols
```

educ
0.098681

A unit increase in educ with all the other terms held constant will be associated with an average change 0.098681 in $\log(wage)$

3.

a.

Hide

```
vars.IV <- c('motheduc','fatheduc','huseduc')
vars.D <- paste(c(vars.Endog,vars.Exog),collapse = " + ")
vars.W <- paste(c(vars.Exog,vars.IV),collapse = " + ")
form.iv <- as.formula(paste("lwage ~",vars.D,"|",vars.W))
fit.iv <- ivreg(form.iv,data = df)
beta.iv <- coef(fit.iv)[vars.Endog]
beta.iv
```

educ
0.05319927

The IV estimate for educ is 0.05319927 < 0.098681

b.

Hide

```
if (beta.ols > beta.iv){
  print('ols is larger with biasd upward')
} else {print('iv is larger with biased downward')}
```

```
[1] "ols is larger with biasd upward"
```

c.

Hide

```
iv.Test <- summary(fit.iv,diagnostics = T)$diagnostics
data.frame(iv.Test['Weak instruments',,drop = F])
```

	df1 <dbl>	df2 <dbl>	statistic <dbl>	p.value <dbl>
Weak instruments	3	417	78.28424	3.476484e-40
1 row				

F statistic is > 10, Instruments are not week.

4. Hausman Test

H_0 : educ is not endogenous

H_1 : educ is endogenous

[Hide](#)

```
data.frame(iv.Test['Wu-Hausman',,drop = F])
```

	df1 <dbl>	df2 <dbl>	statistic <dbl>	p.value <dbl>
Wu-Hausman	1	418	5.119731	0.02416774
1 row				

The p-value < 0.05, reject H_0 . The educ is endogenous.

5. Overidentification Test (Sargan)

$H_0 : E(\mathbf{Z}^\top \varepsilon) = 0$

$H_1 : E(\mathbf{Z}^\top \varepsilon) \neq 0$

[Hide](#)

```
data.frame(iv.Test['Sargan',,drop = F])
```

	df1 <dbl>	df2 <dbl>	statistic <dbl>	p.value <dbl>
Sargan	2	NA	0.4163777	0.8120537
1 row				

The p-value > 0.05, not reject H_0 . Instruments are valid.